Lecture 6: Selecting and analyzing digital evidence

1

Learning Objectives

After studying this chapter, you should be able to:

- Explain and apply structured processes for locating, identifying, and selecting relevant digital evidence, including file categorization, elimination of superfluous data, and deconstruction of file structures.
- Analyze and interpret digital events through event correlation, lead analysis, cloud-based artefact analysis, and the development of accurate forensic timelines.

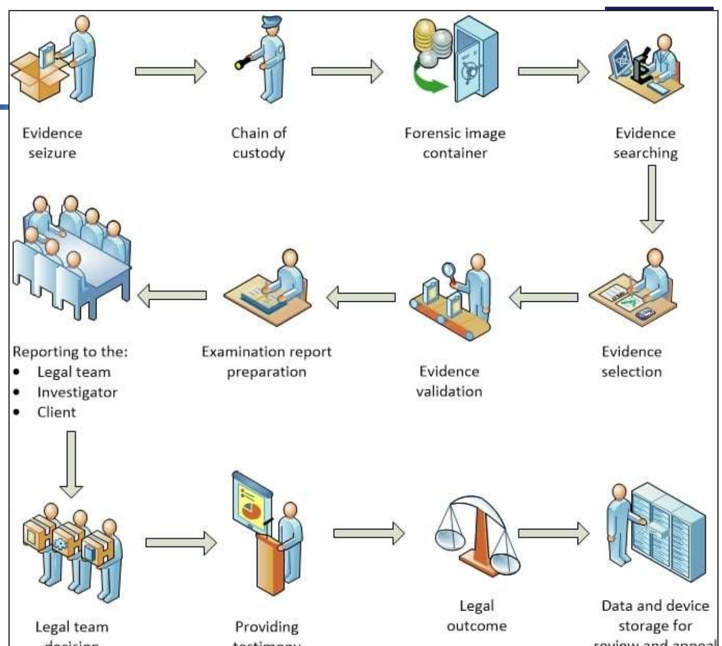
2

Process for Locating & Selecting Digital Evidence

- **Digital forensic models vary**, but most follow a similar structure with iterative stages.
- **Key Phases:**
  - **Recovery & Preservation:** Acquire and secure data (e.g., imaging devices; use of forensic containers like .ASB or .ISK).
  - **Locating Evidence:** Search within recovered data using practitioner knowledge to identify potential items of interest.
  - **Selection & Analysis:** Evaluate discovered items; discard, collect, or tag evidence based on relevance to investigative hypotheses.
  - **Validation:** Confirm the integrity and authenticity of selected evidence.
  - **Presentation:** Prepare findings in reports or courtroom testimony..
- **Benefit of Structured Approach:** Enhances efficiency, completeness, and accuracy compared to unstructured searches.

3

- The figure highlights the examination of digital evidence in the investigative and legal domains.



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Locating Digital Evidence

- Large datasets require filtering to remove system noise and irrelevant data
- Practitioners follow structured processes to focus on relevant investigative material
- **Search methods** target:
  - File names or naming patterns
  - Keywords within content
  - Metadata (e.g., last accessed/modified times)
- Use of **case-specific keyword lists** to extract probative information efficiently
- **Advanced search** options (e.g., in ISeekDesigner):
  - Plain / case-sensitive
  - Whole-word / case-sensitive whole-word
  - Regex / case-insensitive regex
  - Wildcard
  - Hex sequence
  - Beginning-/end-of-word
- Hashing techniques assist in:
  - Matching files to known databases (malware, exploitation material)
  - Identifying altered or renamed files based on content signatures

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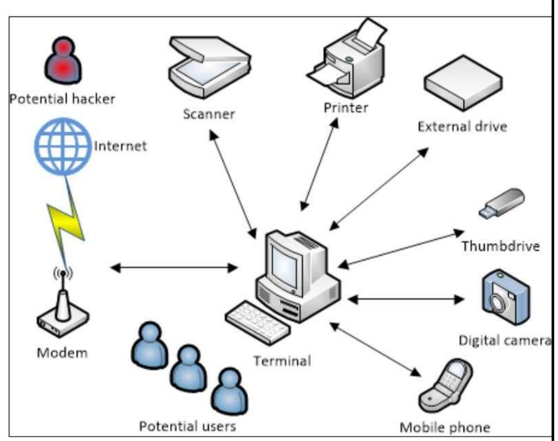
Searching Desktops and Laptops

- A computer terminal connected to the Internet links multiple devices (e.g., scanner, printer, external drives, camera, mobile phone).
- In homes or offices, these setups form larger networks with many users and devices.
- Each connection and device interaction leaves **logs, traces, and records** across the terminal, devices, and Internet services.
- Examples include: e-mail stored on external servers, printer job logs, and data on storage or communication devices.
- These digital traces help **reconstruct events** and provide **supporting evidence** during investigations.



A typical single-terminal network

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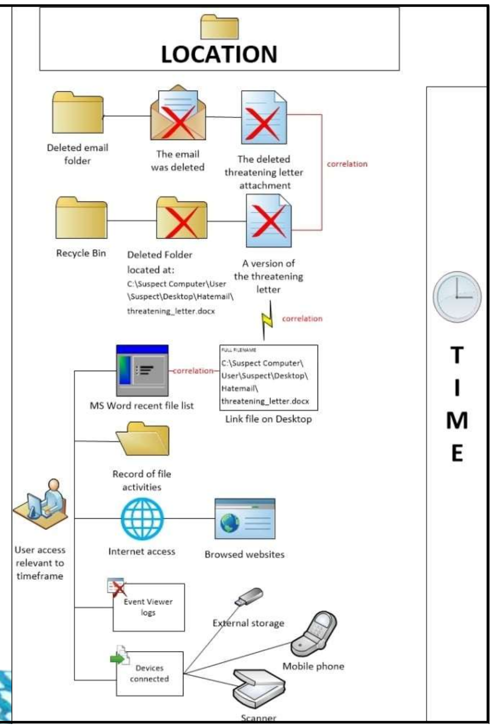
Example

- If we look at the data stored on the terminal, which may be a desktop, laptop, or netbook, traces of relevant data may be located there. In the following diagram, we have recovered a deleted MS Word document from **Recycle Bin**, containing a death threat to another person.
- The intended victim has reported receiving an e-mail message with a Word document attachment from the suspect, whose computer was later seized for examination.
- In this hypothetical scenario (based on an actual case), the practitioner's task is to locate evidence of the e-mail message and attachment. This simple task may reveal the e-mail file and attachment, but it may only locate traces of the message:



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Scenario

- In this scenario, the following occurred and was identified
  - The suspect's email was clean of traces of the e-mail however the deleted e-mail folder held the deleted e-mail and attachment
  - Recycle Bin held a deleted folder and a deleted version of the threatening Word document that was previously located on the computer desktop
  - The Word document file contained metadata relating to the previous location of the file and its recorded author
  - A *link file* (or Jump List log) refers to the location of the file of the same name in this folder that was previously located on the desktop
  - Microsoft Word is installed and shows a record of the creation and modification of a file with the same name in a recent document log
  - User login records show who was logged in at the time of document creation and e-mail activity

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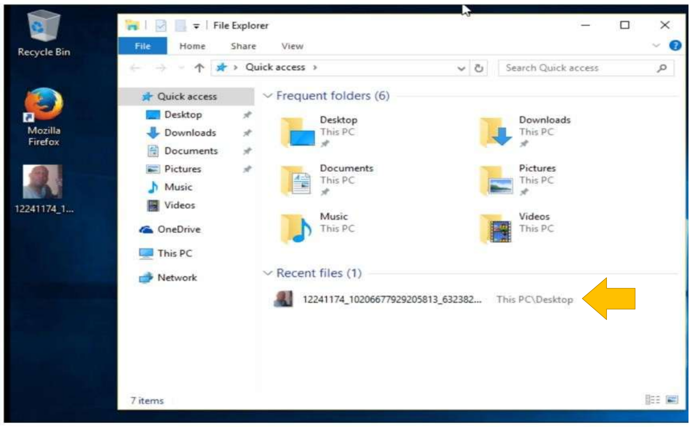
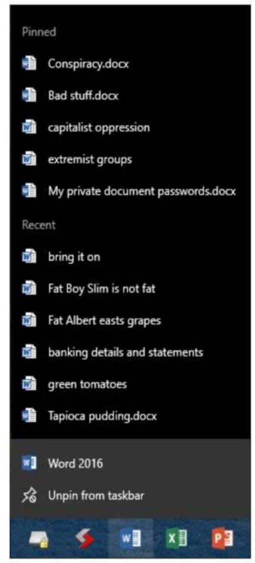
Example of Metadata Trail



The Windows 10 Quick access feature



Word document Jump List showing recent files used by the application on the taskbar

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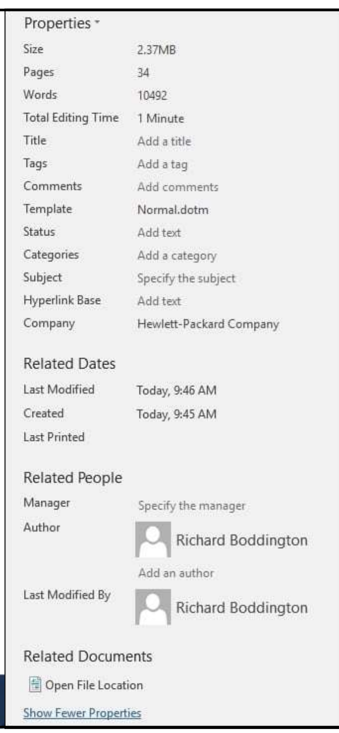

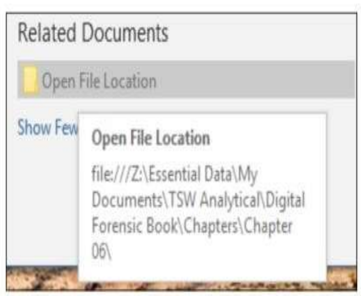

The Related Documents tab-seeking the folder location of a Word document

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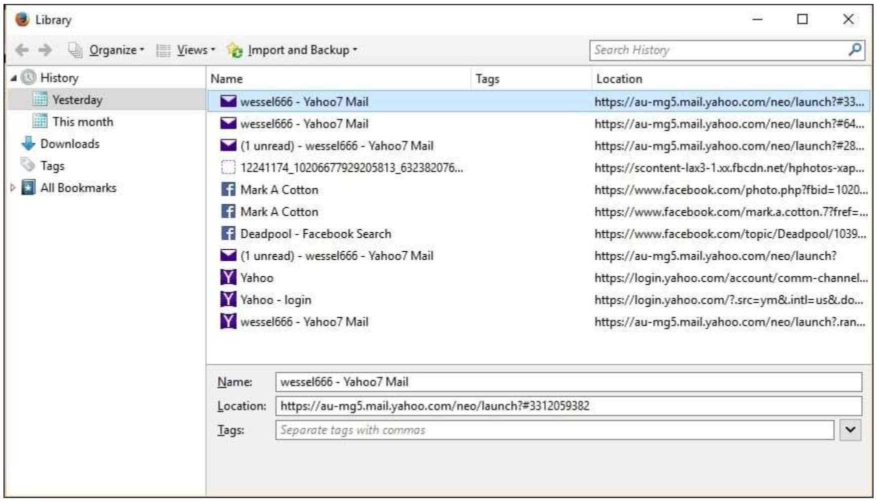

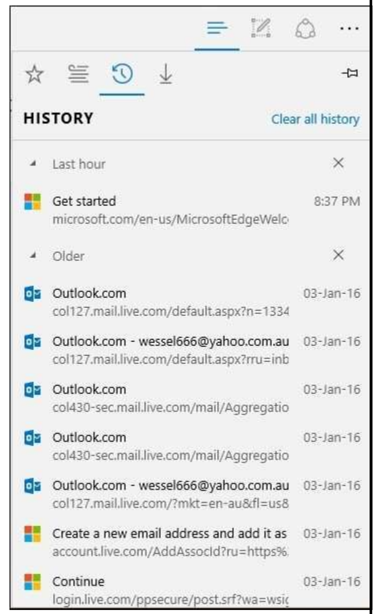

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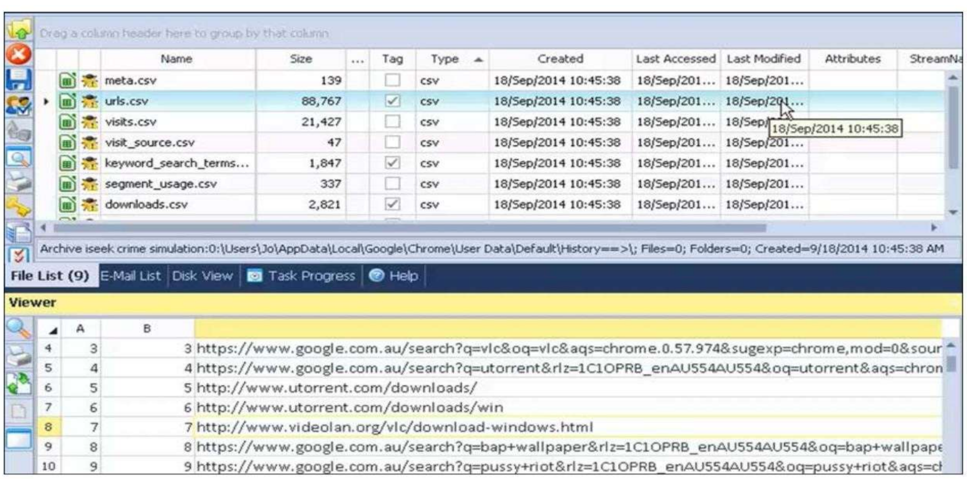
- Specialized tools can often dig into logs/records to extract even more metadata



Spreadsheets and a list of contents extracted from the browser history database

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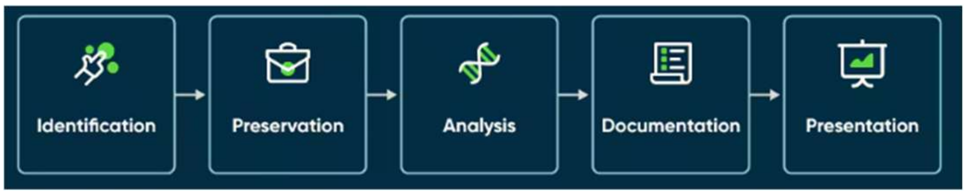
SELECTING DIGITAL EVIDENCE



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Selecting Evidence

- Sometimes referred to as the analysis stage or event reconstruction stage
- Involves analysis of the located evidence to determine what events occurred in the system and their significance and probative value to the case
- Requires practitioners to carefully examine the available digital evidence, ensuring that they do not misinterpret the evidence and make presumptions without carefully cross-checking the information It may be useful to liaise with the investigation and legal teams to ensure that relevant and probative information is selected
- With traditional offenses, the offending act/event is usually manifested, there is a corpse, a theft, or at least a complaint to work with and, usually, a list of potential suspects, based on the simple criteria of: Who knew the victim? Who had access? Who had motive?

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Selecting Evidence

- This approach may not be feasible with digital evidence; e.g. if evidence points towards an Internet based attack, you have well over 700 million suspects online
- It is sometimes problematic to identify the crime during the selection process. Here are some examples:
  - In cybercrime, the nature of the event is often less obvious and immediate e.g. If a hacker steals confidential information, victims may not find out what has been stolen
  - Victims usually rely on being informed by system administrators and may not notice until long after the event has occurred → Identity theft fraud, the fastest-growing financial crime, may take years to be exposed
- Ultimately, the practitioner should consider the quality of the evidence during selection → most digital evidence rarely provides sufficient weight by itself to prove guilt or innocence unless soundly corroborated

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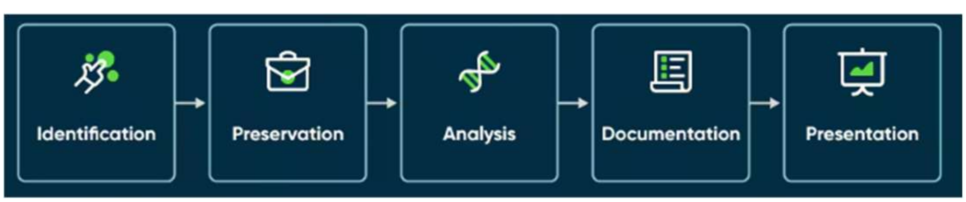
MORE EFFECTIVE FORENSIC TOOLS



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Forensic Tools for Evidence Locating & Selection

- Various forensic tools are now available to assist the practitioner to select and collate data for examination analysis and investigation
- It should be pointed out that in many cases, more than one device is involved, and they require examination as well. This adds to the challenge of the overall forensic examination since no one tool can handle everything
- Available tools for forensics include those for use in
  - Categorizing/eliminating redundant/searching/analyzing files
  - Event/cloud/lead analysis
  - E-mail inspection and analysis
  - Image inspection and analysis
  - Volume shadow copy analysis
- NOTE: textbook references ILook as a tool that can handle all of the above

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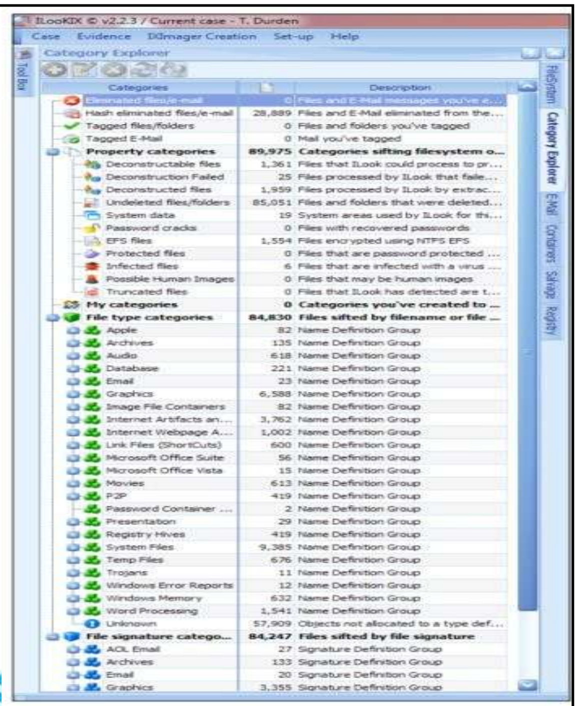
Evidence Locating & Selection

Categorizing files

- Many digital forensic tools used to view and analyze forensic images or attached devices provide helpful user interfaces to locate and categorize information relevant to the examination
- These interfaces have choice to sort, filter, categorize, etc. on varying properties



ILookIX Category Explorer

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Evidence Locating & Selection (Cont.)

Eliminating Superfluous Files

- Computers store many of the same files in duplicate, triplicate, or more versions, which are often scattered in various obscure parts of the system.
- Files with no content or zero-length files also add to the clutter
- Eliminating these files by hiding them from view can significantly reduce the ones that need to be searched manually or by automated applications

File Deconstruction

- Tools can assist in breaking down compound files into smaller parts, e.g. compressed file archives, emails, databases stores etc.
- New tools can perform operations like search, index, extract, etc. without having to actually deconstruct the file until absolutely necessary
- Overall, the tools reduce the amount of time required to examine these compound files as opposed to manual examination

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Evidence Locating & Selection (Cont.)

File Searching

- Indexing is the process whereby chunks of data are catalogued. It is the process of generating a table of text strings that can then be searched almost instantly any number of times.
- The process can take a considerable amount of time to make data available to the examiner depending on operation used, e.g. indexing, hashing, salvaging, etc.

Event Analysis

- Event analysis tools provide graphical representations of events on the subject system → more as a visual aid
- These tools can help map out timelines against various evidence sources or show event sequences to help identify information pertinent to a case

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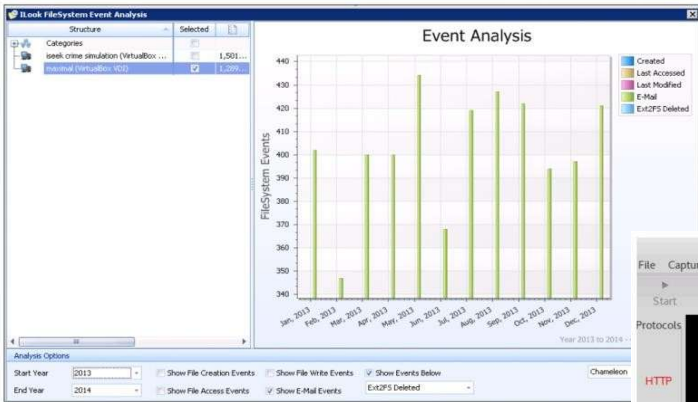
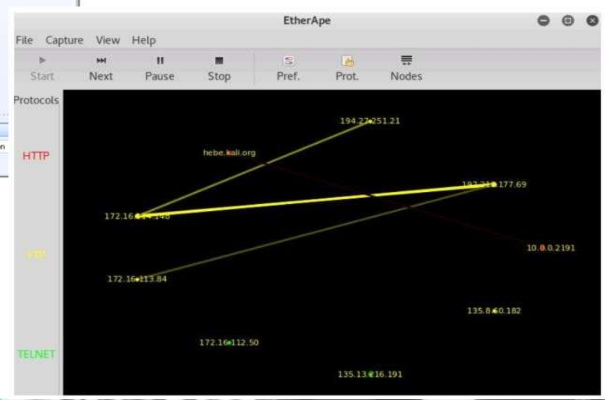


ILookIX Event Viewer showing events between selected dates



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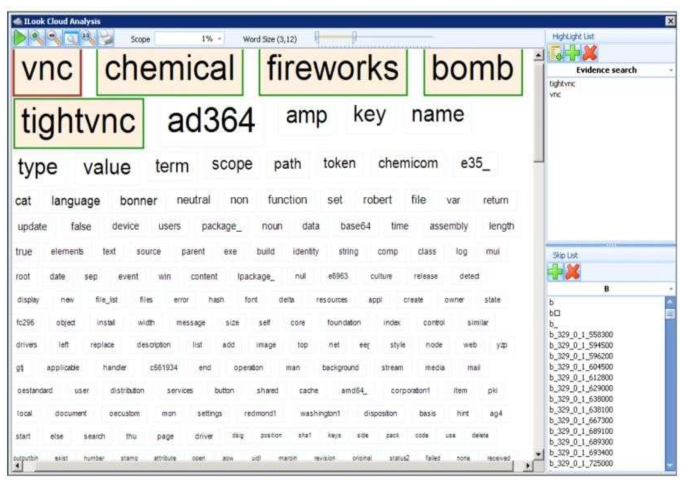
Evidence Locating & Selection (Cont.)

Cloud Analysis Tools

- Another visual representation tool normally used to represent the frequency of terms used in a case
- The analysis highlights to the practitioner important words that might direct the case towards a specific path of resolution



ILookIX Cloud Analysis searching for relevant terms

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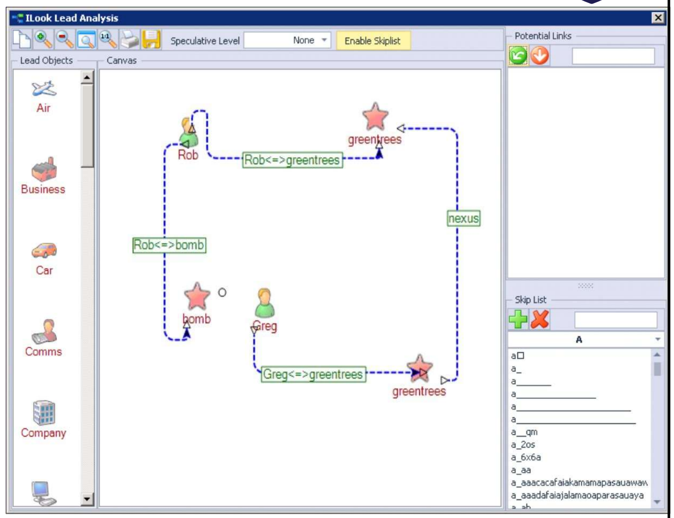
Evidence Locating & Selection (Cont.)

Lead Analysis

- This tool (unique to ILook) helps to discover links within the case data that may not be evident or intuitive and which the practitioner may not be aware of directly or has little background knowledge of to help form relationships manually
- Yet another visual tool with drag-and-drop interface to allow corroboration of evidence from multiple source be done



ILookIX Lead Analysis

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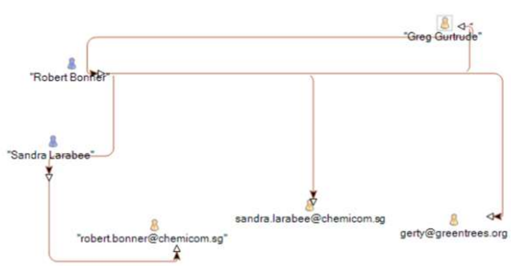
Evidence Locating & Selection (Cont.)

Analyzing E-mail

- Analyzing and selecting evidence from large e-mail datasets is a common task for the practitioner to help discover links between the correspondents within e-mail data → i.e. to trace thread, source, recipient, etc



Scanned Image Authentication

Searching for scanned images is a process that may assist in document authentication or detecting forged documents

iLook has built in function as DetectScannedImages that can then be later be further processed (e.g. OCR, authentication, metadata checking, etc)

My categories	51,967	Categories you've created to ...
[ILookIX] Hash Duplicate Objects	51,902	
[ILookIX] Probable Scanned Images	6	
Dat file web history	58	
Skype	1	

A list of probable scanned images in ILookIX Category Explorer

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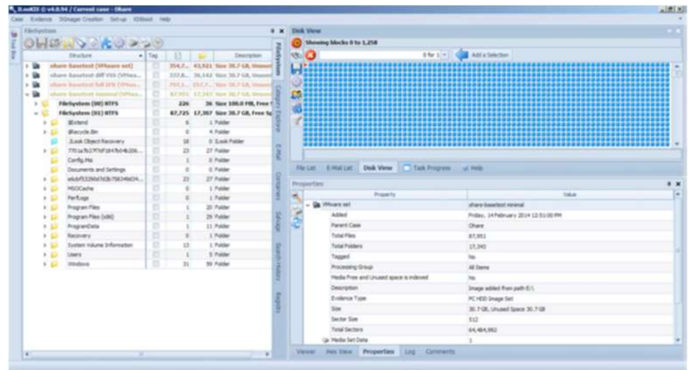
Evidence Locating & Selection (Cont.)

Volume Shadow Copy analysis tools

- A shadow volume, also known as the Volume Snapshot Service (VSS), is a service that creates point-in-time copies of files.
- VSS recovery is a method of recovering existing and deleted files from the volume snapshots available on the system →a valuable resource for locating previously unavailable data to assist an investigation



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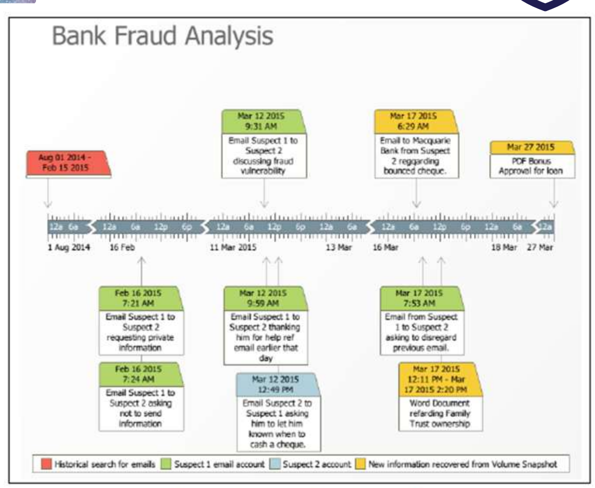
Evidence Locating & Selection (Cont.)

Timelines and other analysis tools

- Timelines help reconstruct the chronology of events over a specific period.
- They can become overcrowded and difficult to read when covering long or detailed periods.
- It is useful to design timelines that are more reader-friendly.
- A simplified main chart can present only the key events in a clean, uncluttered way.
- Major events can be expanded into separate, detailed charts.
- These detailed charts should include links to related or neighboring events for continuity.



Timeline of events using Timeline Maker Pro

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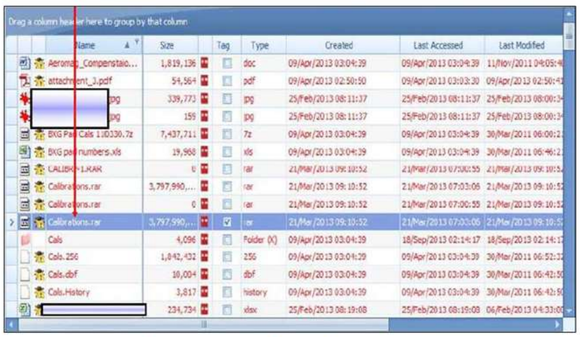
Case study – illustrating the recovery of deleted evidence held in volume shadows

The case study involves an emerging business providing specialist technical services to a range of mining exploration companies. The business had lost a number of its technical experts to a rival business and noticed a drop in in its market share. There was a suspicion that departing staff members may have stolen some proprietary knowledge and expertise, which they later shared with the competing business. A forensic examination of the business' server and the computer terminals used by the departed staff members was initiated to determine whether some form of industrial espionage or sabotage had occurred.

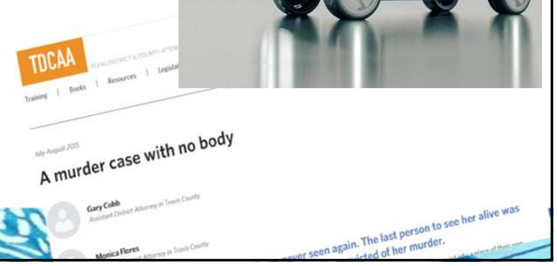


Drag a column header here to group by that column

Name	Size	Tag	Type	Created	Last Accessed	Last Modified
Aeromap_Compenstaio...	1,819,136		doc	09/Apr/2013 03:04:39	09/Apr/2013 03:04:39	11/Nov/2011 04:05:4
attachment_3.pdf	54,564		pdf	09/Apr/2013 02:50:50	09/Apr/2013 03:03:30	09/Apr/2013 02:50:4
[illegible]jpg	339,773		jpg	25/Feb/2013 08:11:37	25/Feb/2013 08:11:37	25/Feb/2013 08:00:3
[illegible]jpg	159		jpg	25/Feb/2013 08:11:37	25/Feb/2013 08:11:37	25/Feb/2013 08:00:3
BXG Par Cals 110330.7z	7,437,711		7z	09/Apr/2013 03:04:39	09/Apr/2013 03:04:39	30/Mar/2011 06:00:2
BXG par numbers.xls	19,968		xls	09/Apr/2013 03:04:39	09/Apr/2013 03:04:39	30/Mar/2011 06:46:2
CALIBR~1.RAR	0		rar	21/Mar/2013 09:10:52	21/Mar/2013 07:00:55	21/Mar/2013 09:10:5
Calibrations.rar	3,797,990,...		rar	21/Mar/2013 09:10:52	21/Mar/2013 07:03:06	21/Mar/2013 09:10:5
Calibrations.rar	0		rar	21/Mar/2013 09:10:52	21/Mar/2013 07:00:55	21/Mar/2013 09:10:5
Calibrations.rar	3,797,990,...	☑	rar	21/Mar/2013 09:10:52	21/Mar/2013 07:03:06	21/Mar/2013 09:10:5
Cals	4,096		Folder (X)	09/Apr/2013 03:04:39	18/Sep/2013 02:14:17	18/Sep/2013 02:14:1
Cals.256	1,842,432		256	09/Apr/2013 03:04:39	09/Apr/2013 03:04:39	30/Mar/2011 06:52:3
Cals.dbf	10,004		dbf	09/Apr/2013 03:04:39	09/Apr/2013 03:04:39	30/Mar/2011 06:42:5
Cals.History	3,817		history	09/Apr/2013 03:04:39	09/Apr/2013 03:04:39	30/Mar/2011 06:42:5
[illegible]	234,734		xlsx	25/Feb/2013 08:19:08	25/Feb/2013 08:19:08	06/Feb/2013 04:33:0

Recovered data from a Dropbox folder held in a volume shadow snapshot

Refer to: Detail Case Study



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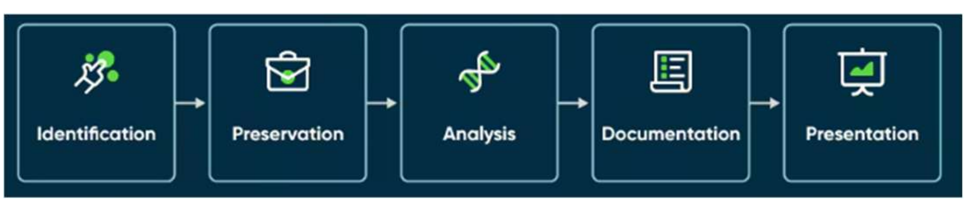
DIGITAL FORENSIC OPERATING SYSTEMS



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Comparison of Digital Forensics Operating Systems

- Both commercial and open-source forensic tools exist; many open-source options are Linux-based and included in forensic distributions.
- Key priority for any tool: **preservation and integrity of evidence**, regardless of licensing model.
- Commercial tools are often robust and user-friendly but may cost **thousands of dollars**.
- Open-source tools are **free**, widely reviewed, community-supported, and transparent due to non-proprietary code.
- Many open-source forensic distros are available, each offering a suite of tools for investigators of all levels.
- This section focuses on the forensic capabilities provided by **Kali Linux**, with other popular distros also introduced.

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Computer Aided INvestigative Environment (CAINE)

- CAINE is a live-response forensic Linux distribution designed for incident response and digital investigations.
- It supports multiple boot options: Safe Mode, Text Mode, Live System, and Run in RAM.
- The latest release, CAINE 14.0 (Lightstream), is live-only and cannot be installed on a hard drive.
- Can be download at https://www.caine-live.net/

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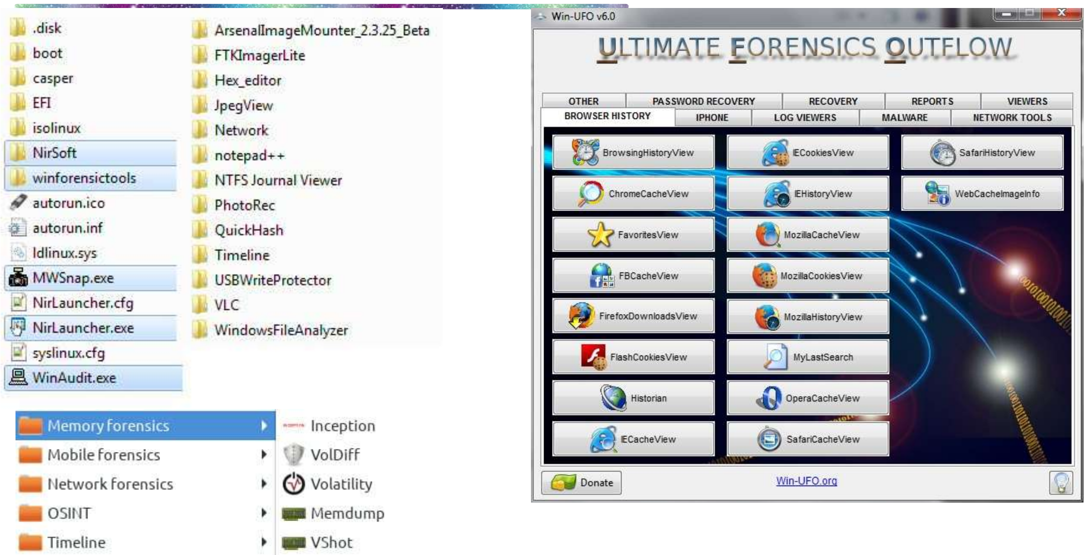
Computer Aided INvestigative Environment (CAINE)



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CSI Linux

- CSI Linux is a modern DFIR and blue-team–focused distribution created by Jeremy Martin and his team.
- It provides a powerful environment for digital forensics, threat hunting, incident response, and OSINT, and is often used alongside Kali Linux.
- CSI Linux is available as a virtual appliance for VirtualBox and VMware, and also as a bootable live image for field forensics.
- The developers are also working on a dedicated CSI Linux SIEM to further enhance the platform's defensive capabilities.
- The distribution can be downloaded from the official site: csilinux.com/download.

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CSI Linux

Key Features

Comprehensive OSINT Tools

A full suite of OSINT tools for gathering intelligence from public sources efficiently and securely.

Malware Analysis

Tools designed for analyzing and understanding malicious code in a safe environment.

Incident Response

Quick and effective response tools to tackle security incidents and reduce potential damage.

Integrated Tools

Network Forensics

Analyze network traffic, identify anomalies, and reconstruct events.

Digital Evidence Management

Organize and manage digital evidence with ease using built-in case management tools.

Data Recovery

Advanced tools to recover data from damaged or corrupted media.

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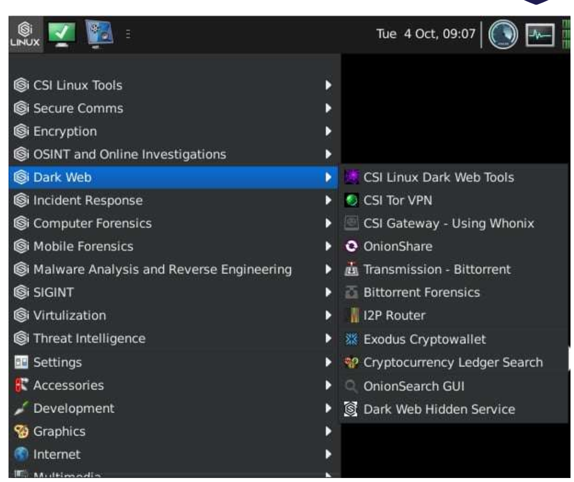
CSI Linux



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Kali Linux

- **Origins:** Evolved from BackTrack into Kali Linux (2015); widely used by penetration testers and security enthusiasts.
- **Use in Education:** Star tool in CEH courses due to extensive bundled security programs (scanning, reconnaissance, exploitation, reporting).
- **Forensic Capability:** Can be used for live-response investigations with tools that preserve evidence integrity.
- **Full OS Functionality:** Supports full installation on hard drives or flash drives; includes productivity tools, hardware drivers, and runs on 32-bit & 64-bit systems.
- **Mobile Compatibility:** Installable on certain mobile devices (e.g., Nexus, OnePlus).
- **Live Forensic Mode:** Ensures original evidence is untouched, disables auto-mounting of drives, and preserves integrity.
- **Versatility:** Combines penetration testing and forensic tools in one platform, ideal for **purple teaming**

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Kali Linux

- Download Link: https://www.kali.org/get-kali/#kali-platforms
- Collection of tools: https://www.kali.org/tools/

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Summary

What was covered today:

- Described in more detail the process of locating and selecting evidence in terms of a general process and also further explained the nature of digital evidence
- Introduction to various advanced analysis and recovery tools that can speed up and make more efficient the evidence location and selection processes

Reference Source: Practical Digital Forensics (Packt Publishing)

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THANKS – Q&A

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